



Module 4: Implementing Sentiment Analysis

AWS Academy Natural
Language Processing

Module overview

Sections

- Section 1: Introducing the scenario
- Section 2: Identifying the steps for text processing
- Section 3: Examining the algorithms for sentiment analysis
- Section 4: Discussing and walking through the lab solution

Lab

- Challenge Lab: Implementing Sentiment Analysis

Module objectives

- At the end of this module, you should be able to:
- Describe the challenges in working with social media data
- Evaluate machine learning (ML) algorithms and tools that are used in natural language processing (NLP)
- Create a solution to a sentiment analysis business problem



Section 1: Introducing the scenario

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Business case

- Movie review – Use the text of the review to calculate:
 - + (positive sentiment) or - (negative sentiment) ratings
 - Count of positive and negative reviews
- Labeled dataset that contains 50,000 reviews

Dataset

Text	Sentiment
This is a charming movie starring everyone's favorite cartoon chipmunks. In this feature we follow the band of rodents on an unforgettable balloon race around the world. Although there are lows, including an orphan penguin, all in all it's a great family film.	1
I really should have learned more about this movie before renting it. It was one of those movies where you keep watching it figuring it's got to get better. Then, when it ends, you feel stupid for having wasted precious time in your life that you can never get back. The pictures of the shuttle looks like it was done with a little toy inside of a box and the spacewalking scenes were funny because you could see the strings attached to the space suits. The script was lacking and the car chase scene with the guy bleeding and going unconscious was incredible because he drove better than I could have on one of my best days. All in all, I have seen worse but this sure isn't one I'd recommend or want to remember.	0



Section 2: Identifying the steps for text processing

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Text-classification steps

Collect labeled dataset



Normalize the data



Transform text into feature vectors



Train a classifier



Use metrics to tune model

What processing steps might you take?

- "Diego "“leader of debonair”" Ramirez plays the Big Boss (IVAN) who preys upon the unfortunate John Stiles (PINKY) by forcing the hapless ex-con to exploit his ill-found new position in a bank. Nikki Wolf (Miss PELHAM) most effectively provides the female interest, whom Pinkie simply cannot resist.

It seems they were unable to decide on one name for this film so instead they used four makes sense ???

Sadly, this turned out to be one of Niven's last roles.

Overall, this film is fun and well worth watching if you manage to catch one of its rare or late night TV screenings.

"

Handling informal text data

- Informal text data:
 - Markup
 - Non-text data and emojis
 - Shortcuts – ‘s, ‘re, ‘r
 - URLs
 - Word spacing
 - Non-standard spelling
 - No grammar
 - Multilingual
 - Transliteration
 - Homonyms
- Handling sarcasm, trolls, memes, slang, and fake news

Processing steps

- Cleaning up text:
 - Convert to lowercase
 - Remove extra spaces
 - Remove HTML tags
- Remove stopwords
 - Exception list
- Tokenize or vectorize text

Handling missing data

- Can you fill in missing values?
- If the missing data is a small amount—drop the data
- Use standard missing-data practices for other features

Section 2: Summary

- ML steps
 - Gather the dataset
 - Normalize the text
 - Transform the text into features
 - Train a model
 - Use metrics to tune
- Handle informal text data
- Perform processing steps
- Handle missing data



Section 3: Examining the algorithms for sentiment analysis

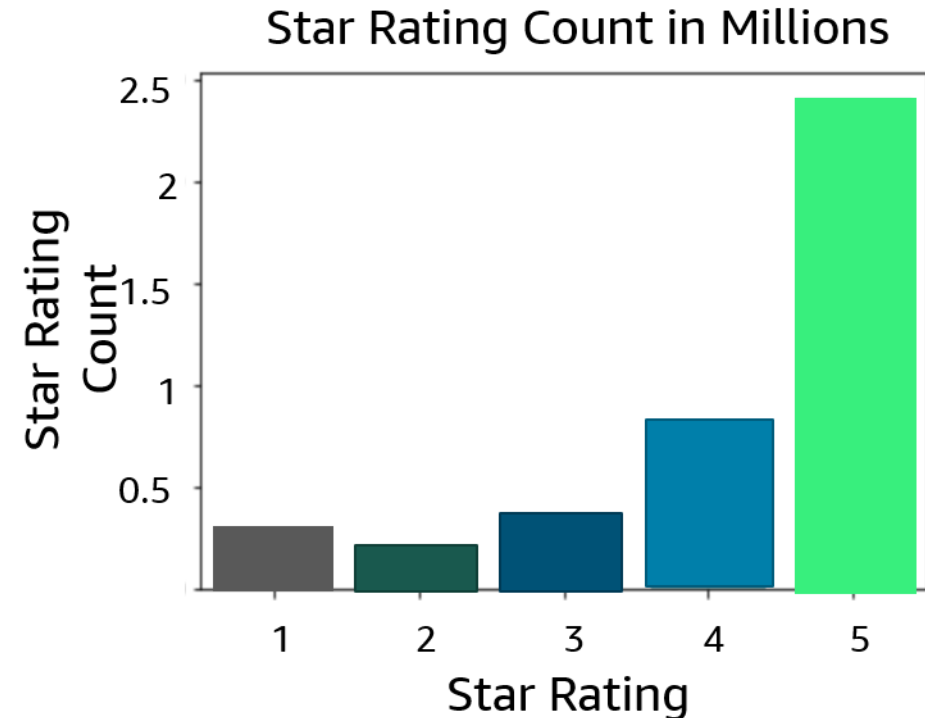
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What kind of ML problem is it?

- Supervised
- Binary classification
- Goals are to maximize correct answers
 - Metrics: Error, accuracy

Class imbalance

- Number of samples per class is not equally distributed
- The ML model might not work well for the infrequent classes
- Examples:
 - Fraud detection
 - Anomaly detection
 - Medical diagnosis



Amazon review dataset: The number of five-star reviews almost equals the total of the other four types of star reviews combined

Addressing the class imbalance problem

Downsampling

- Reduce the size of the dominant or frequent classes.

Upsampling

- Increase the size of the rare or small classes.

Data generation

- Create new records— similar, but not identical.
- For example, create similar images that distort original images by rotating, introducing noise, or skewing.

Sample weights

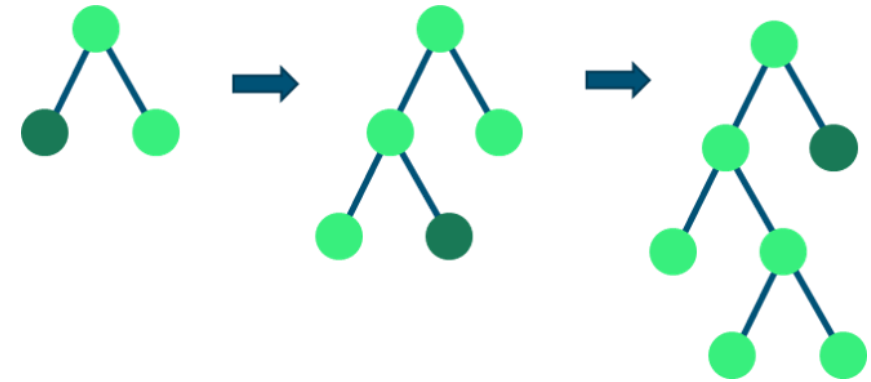
- For a model that uses a cost function, assign higher weights to rare classes and lower weights to dominant classes.

Classifier performance

- Sparse vectors
 - Vector has a large number of zeros
- Hyperparameter tuning
 - Is important to allocate time for hyperparameter tuning, which can improve the results of a model
 - Understand the metrics and tunable parameters for the algorithm that you are using

XGBoost

- Is an open-source implementation of the gradient boosted trees algorithm
- Has performed well in ML competitions
- Robustly handles various data types, relationships, distributions, and a large number of hyperparameters



BlazingText classification

- BlazingText classification
 - Sentiment analysis
 - Spam detection
 - Hashtag prediction
- Training data format

```
__label__0 this movie is the worst...  
__label__1 One of the best movies..
```

- Recommendations for Amazon Elastic Compute Cloud (Amazon EC2) instances –
 - <2 GB – C5 instance
 - >2 GB – ml.p2 or ml.p3 – Single Graphics Processing Unit (GPU) instances

Using pretrained models

- Natural Language Toolkit (NLTK)
 - Vader lexicon – Uses bag-of-words (BOW) approach
- Textblob
 - Uses BOW classifier
 - Includes subjectivity analysis
- Flair
 - Based on character-level, long short-term memory (LSTM) neural network
 - Factors sequences of letters and words
 - Can handle out-of-vocabulary (OOV) words

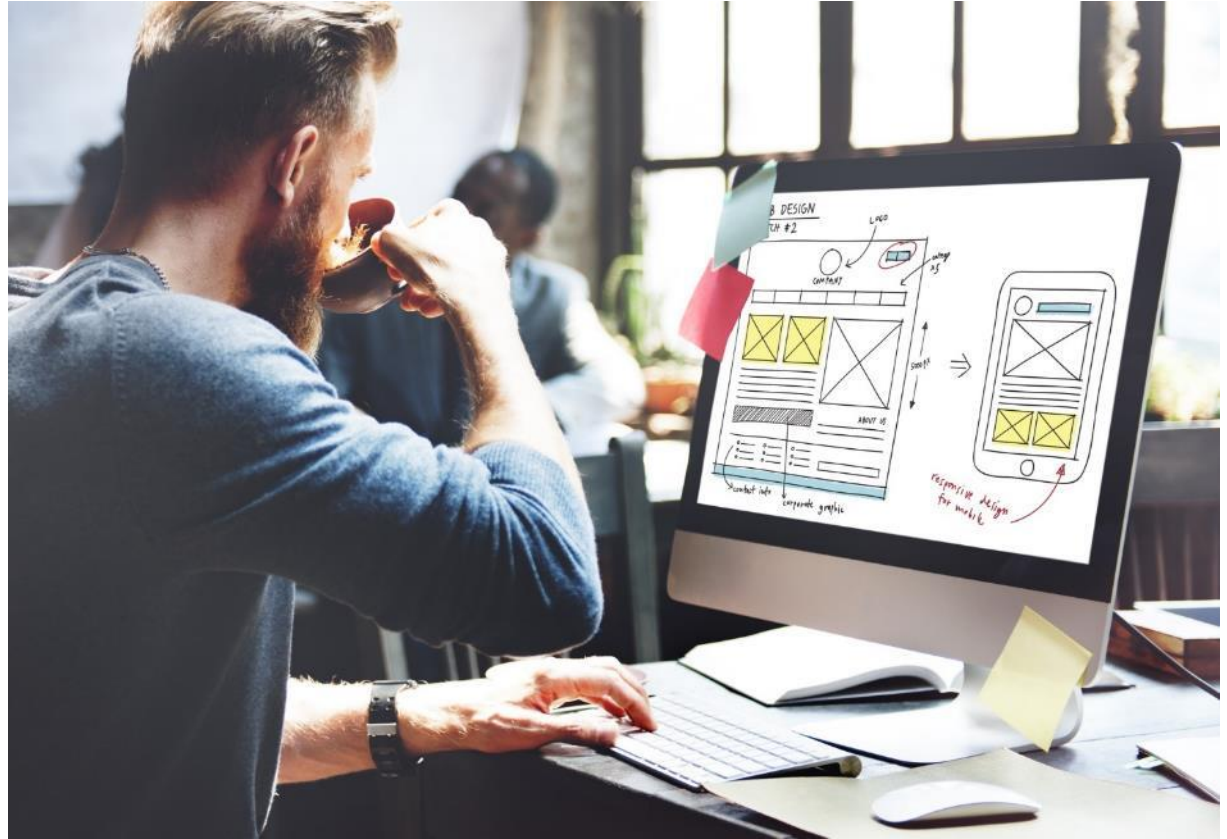
Amazon Comprehend

- Fully managed service
- Includes application programming interface (API) for sentiment analysis
 - No training required
- Can train a custom classifier
 - Train and test as you would an Amazon SageMaker algorithm
 - Training data must be in correct format
 - Predictions include the class and probability score

Section 3: Summary

- ML problem
- Class imbalance
- Built-in Amazon SageMaker algorithms
 - XGBoost
 - BlazingText
- Pretrained models
- Amazon Comprehend

Module 4 Challenge Lab: Implementing Sentiment Analysis





Section 4: Discussing and walking through the lab solution

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Lab summary

	First pass	Second pass	Third pass	Fourth pass	Fifth pass
Algorithm or service	XGBoost	XGBoost	XGBoost	BlazingText	Amazon Comprehend
Vectorizer	CounterVector	CounterVector	CounterVector	Word2Vec	n/a
Text processing steps	None	Normalized Stopword removed	Normalized Stopword removed	None	n/a
Hyperparameter tuned (Yes or No)			YES		
~Accuracy	82.22%	51%	85%	87%	88%

Note: Your results will vary from these results.

Questions?

- What might you do differently if you had more time?
- How might you improve the results?



Thank you

Corrections, feedback, or other questions?

Contact us at <https://support.aws.amazon.com/#/contacts/aws-academy>.